

Vision Transformer With Contrastive Learning for Hyperspectral Image Classification

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Abstract—The vision transformer (ViT) has become a hot topic in image processing due to its global feature extraction capabilities. However, the ViT suffers from over-smoothing in feature extraction and over-fitting in the training procedure, so it is hard to achieve satisfactory performance in hyperspectral image (HSI) classification. To address these issues, we propose a ViT with contrastive learning (CViT). The network architecture includes a patch embedding module, transformer blocks, and a classifier. The training of CViT can be considered as an optimization problem with a supervised contrastive loss, an unsupervised contrastive loss, and an ℓ_1 -regularizer with respect to linear self-attention weights. Specifically, the supervised contrastive loss is proposed to alleviate the negative effects of HSI features' spectral variability and spatial diversity by increasing intraclass consistency. On the other hand, the unsupervised contrastive loss is exploited to reduce redundancy by reconstructing global structural information. In particular, regularized linear self-attention weights reduce the over-smoothing issue. Extensive experimental results on three HSI datasets demonstrate that the proposed CViT achieves competitive performance.

Index Terms—Contrastive learning, hyperspectral image (HSI) classification, self-attention mechanism, vision transformer (ViT).

I. INTRODUCTION

HYPERSPECTRAL images (HSIs) can accurately characterize the properties of ground objects, constituting a 3-D data cube containing hundreds of narrow and continuous bands [1]. As a vital application of remote sensing, HSI classification aims to assign a category label to each HSI pixel and has been widely applied in geological surveys, precision agriculture, urban planning, and so on [2]. HSI classification is a challenging task due to the spectral variability caused by environmental factors and the difficulty of obtaining HSI labels.

In the past decades, researchers have proposed a large number of HSI classification methods, where spatial-spectral representation-based methods have become a mainstream topic. Among them, manual feature engineering is applied to extract features, and then, machine learning classifiers

are utilized for pixelwise classification. These methods have profoundly impacted the extraction of HSI features, but they are limited by the manual extraction of features and the shallowness of the obtained features. Deep learning-based methods have the representation power to obtain deeper and learnable features via an end-to-end automated extraction procedure [3], [4]. There is a large range of classic network architectures for HSI classification in the literature [5]. Convolutional neural networks (CNNs) can extract features from shallow to deep layers by stacking convolutional kernels [6]. Moreover, CNNs have shown great potential for extracting spatial features and have dominated HSI classification tasks.

Recently, transformer-based methods for modeling sequence data have attracted increasing interest [7]. In general, the core modules of transformers consist of multihead self-attention mechanisms (MSAs), residual connections, and feed-forward multilayer perceptrons (MLPs) [8]. Vision transformers (ViTs) have achieved remarkable success in many computer vision tasks, where images are separated into multiple patches and embedded to construct a sequence of tokens [9]. Considering the application of ViT for pixelwise classification of HSIs [10], the advantage is that ViT has a global receptive field, so that long-distance dependencies between pixels can be obtained. Huang et al. [11] proposed a 3-D-Swin-transformer-based contrastive learning method to mine the pixel-level representations. ViT overcomes the shortcomings of CNNs, where the small scales and fixed shapes of convolutional kernels lead to limitations in feature extraction.

However, there are still some challenges for ViTs in extracting discriminative and generalized features. First, dense connections and high-dimensional computation of ViTs cause the problem of over-smoothing, exhibiting the strong correlation between tokens in the extracted features [12]. The over-smoothing phenomenon reduces the interclass variability of the features and leads to a decrease in discriminative power. Second, ViTs have many parameters with high degrees of freedom and require a large amount of data for training [13]. But, HSIs have limited labeled samples, and it is easy to fall into over-fitting and local optima when training a ViT for HSI classification.

To address these issues, we propose a ViT with contrastive learning (CViT) to achieve high accuracy and generalizability in HSI classification, as shown in Fig. 1. The computation of the MSA is simplified as linear mappings, and a regularization term is introduced to constrain the degrees of freedom of

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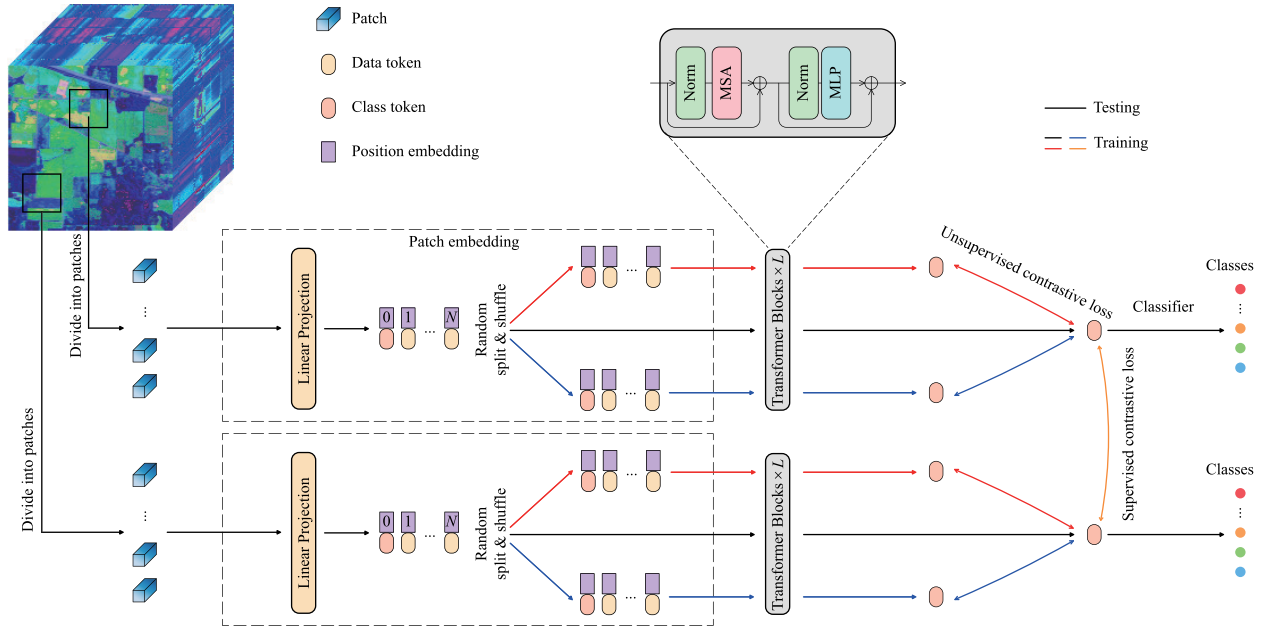


Fig. 1. Overview of the CViT. The architecture of CViT contains a patch embedding module, transformer blocks, and a classifier. First, the input data are divided into patches and then embedded into a sequence of tokens. To better train the model, we obtain different views of the same sample by randomly splitting and shuffling to construct unsupervised contrastive learning. Moreover, the features of different samples constitute supervised contrastive learning. Finally, an ℓ_1 -regularizer is utilized to alleviate the over-fitting issue of self-attention mechanisms.

TABLE I
TECHNIQUES ADOPTED IN THE PROPOSED METHOD
AND THE COMPARISON METHODS

Methods	Self-attention	Supervised contrastive loss	Unsupervised contrastive loss	Regularizer
[9]	Vanilla	×	×	×
[10]	Vanilla	×	×	×
[11]	Shift window	×	✓	×
[14]	Shift window	×	×	×
[15]	×	×	✓	×
[16]	×	×	✓	×
[17]	×	✓	×	×
[18]	×	×	✓	×
Ours	Linear mapping	✓	✓	✓

the parameters. Moreover, we introduced contrastive learning that can mine the internal information of data without labeling information to provide additional supervised information. Table I shows the techniques adopted in the proposed method and the comparison methods. Finally, extensive experiments conducted on three widely used HSI datasets show that the proposed CViT exhibits competitive performance with limited training samples. The main contribution of this letter can be summarized as follows.

- 1) We simplify the MSA and apply an ℓ_1 -regularizer on MSA weights to alleviate the over-smoothing issue and improve the feature representation capability of CViT.
- 2) Supervised contrastive learning is exploited to reduce intraclass feature differences and increases interclass feature differences.
- 3) The proposed unsupervised contrastive learning allows CViT to capture global structural information and increase the generalization of features.

II. PROPOSED METHOD

A. Architecture of the CViT

1) *Patch Embedding for HSI*: The patch embedding is employed to reduce the dimension of the input data $\mathcal{X} \in \mathbb{R}^{C \times H \times W}$ and map $\mathcal{X}$ into a sequence of tokens $\mathbf{Z}^{(0)} = [\mathbf{z}_n^{(0)}]_{n=0}^N$, where C is the number of channels, $H \times W$ represents the band image size, $\mathbf{z}_n^{(0)} \in \mathbb{R}^D$, D is the dimension of tokens, and $(N + 1)$ is the number of tokens. To obtain the tokens, we first divide $\mathcal{X}$ into a vectorized form collection of patches $\mathbf{X} = [\mathbf{X}_n]_{n=1}^N$, where $\mathbf{X}_n \in \mathbb{R}^{C \times P^2}$, $P \times P$ is the size of patches, and the number of patches $N = HW/P^2$. $\mathbf{X}_n$ can be written as $\mathbf{X}_n = [\mathbf{x}_{n,c}]_{c=1}^C$, where $\mathbf{x}_{n,c} \in \mathbb{R}^{P^2}$. Then, learnable parameters $\mathbf{E}_c \in \mathbb{R}^{P^2 \times D}$ and position embeddings $\mathbf{E}_n^{(\text{pos})} \in \mathbb{R}^D$ ($n = 0, 1, \dots, N$) are utilized to embed the patches into tokens by the following equation:

$$\mathbf{z}_n^{(0)} = \sum_{c=1}^C \mathbf{x}_{n,c} \mathbf{E}_c + \mathbf{E}_n^{(\text{pos})}, \quad n = 1, 2, \dots, N. \quad (1)$$

We define $\mathbf{z}_0^{(0)} = [1, 1, \dots, 1]^T + \mathbf{E}_0^{(\text{pos})}$ as an extra class token, and all the tokens are collected into $\mathbf{Z}^{(0)} = [\mathbf{z}_n^{(0)}]_{n=0}^N$ in the 0th block. Similarly, the tokens in the ℓ th block can be written as $\mathbf{Z}^{(\ell)} = [\mathbf{z}_n^{(\ell)}]_{n=0}^N \in \mathbb{R}^{(N+1) \times D}$, where $\ell = 1, 2, \dots, L$, and L is the number of transformer blocks.

2) *Multihead Self-Attention Mechanism*: The MSA is utilized to fuse global information based on the similarity of tokens. To calculate the MSA, the tokens $\mathbf{Z}^{(\ell)}$ in the ℓ th block are first mapped to the *Queries* $\mathbf{Q}_m^{(\ell)} \in \mathbb{R}^{(N+1) \times D}$, *Keys* $\mathbf{K}_m^{(\ell)} \in \mathbb{R}^{(N+1) \times D}$, and *Values* $\mathbf{V}_m^{(\ell)} \in \mathbb{R}^{(N+1) \times D}$ for the m th head ($m = 1, 2, \dots, M$), where M denotes the number of heads

$$\begin{aligned} \mathbf{Q}_m^{(\ell)} &= \mathbf{Z}^{(\ell)} \mathbf{U}_m^{(\ell, Q)}, \mathbf{K}_m^{(\ell)} = \mathbf{Z}^{(\ell)} \mathbf{U}_m^{(\ell, K)}, \\ \mathbf{V}_m^{(\ell)} &= \mathbf{Z}^{(\ell)} \mathbf{U}_m^{(\ell, V)}, \quad m = 1, 2, \dots, M \end{aligned} \quad (2)$$

where $\mathbf{U}_m^{(\ell,Q)}$, $\mathbf{U}_m^{(\ell,K)}$, and $\mathbf{U}_m^{(\ell,V)}$ are learnable parameters. Then, we adopt the similarity between the *Queries* and *Keys* to calculate the MSA by the following equation:

$$\mathbf{A}_m^{(\ell)} = \mathbf{Q}_m^{(\ell)} \mathbf{K}_m^{(\ell)T} / \tau, \text{MSA}(\mathbf{Z}^{(\ell)}) = \sum_{m=1}^M \tilde{\mathbf{U}}_m^{(\ell)} \mathbf{A}_m^{(\ell)} \mathbf{V}_m^{(\ell)} \quad (3)$$

where $\mathbf{A}_m^{(\ell)} \in \mathbb{R}^{(N+1) \times (N+1)}$ stands for MSA weights, the learnable parameter $\tilde{\mathbf{U}}_m^{(\ell)} \in \mathbb{R}$, and $\tau \in \mathbb{R}^+$ is a temperature parameter. Note that, we obtain $\mathbf{A}_m^{(\ell)}$ directly by linear computation to improve computational efficiency rather than by softmax($\cdot$) mapping used in ViT [9].

3) *Multilayer Perceptron*: The MLP operation is utilized to further abstract the features on the channel by the following equation:

$$\text{MLP}(\mathbf{Z}^{(\ell)}) = \mathbf{Z}^{(\ell)} \mathbf{H}^{(\ell)} + \mathbf{b}^{(\ell)} \quad (4)$$

where the learnable parameters $\mathbf{H}^{(\ell)} \in \mathbb{R}^{D \times D}$ and $\mathbf{b}^{(\ell)} \in \mathbb{R}^D$.

4) *Transformer Block*: The transformer block includes the layer normalization (Norm($\cdot$)) [9], the MSA, the MLP, and residual connections calculated by the following equation:

$$\begin{aligned} \tilde{\mathbf{Z}}^{(\ell)} &= \mathbf{Z}^{(\ell)} + \text{MSA}(\text{Norm}(\mathbf{Z}^{(\ell)})) \\ \mathbf{Z}^{(\ell+1)} &= \tilde{\mathbf{Z}}^{(\ell)} + \text{MLP}(\text{Norm}(\tilde{\mathbf{Z}}^{(\ell)})). \end{aligned} \quad (5)$$

B. Optimization

In this section, we discuss the optimization objectives of the proposed CViT model. In addition to the basic cross-entropy loss for classification, we introduce supervised contrastive loss and unsupervised contrastive loss to improve the accuracy and generalization of the model. Furthermore, the ℓ_1 -regularizer is imposed on the MSA weights to improve the generalizability of the model. For simplify, let Θ and $\mathcal{A} = [\mathbf{A}_m^{(\ell)}]_{m=1, \ell=1}^{M,L}$ denote a vectorized form collection of all parameters and MSA parameters in the proposed CViT, respectively.

1) *Cross-Entropy Loss*: Let $\hat{\mathbf{y}} = [\hat{y}_1, \hat{y}_2, \dots, \hat{y}_j, \dots, \hat{y}_J]$ denote the prediction of the proposed CViT, where $\hat{\mathbf{y}} = \text{MLP}_{\text{cls}}(\mathbf{z}_0^{(L)})$, and J is the number of classes. The cross-entropy loss term is calculated by the following equation:

$$\mathcal{L}_{\text{CE}}(\Theta) = -\frac{1}{J} \sum_{j=1}^J y_j \log(\text{softmax}(\hat{y}_j)) \quad (6)$$

where $y_j \in \{0, 1\}$ (if the input training sample belongs to the j th class, $y_j = 1$; otherwise, $y_j = 0$).

2) *Supervised Contrastive Loss*: Given a mini-batch consisting of I samples, let $\mathbf{z}_0^{(L)}(i)$ ($i \in [1, I]$) denote the class token corresponding to the i th sample. The class token $\mathbf{z}_0^{(L)}(s)$ ($s \in [1, S]$) corresponding to the s th sample belongs to the same class with the i th sample, while the class token $\mathbf{z}_0^{(L)}(t)$ ($t \in [1, T]$) corresponding to the t th sample belongs to any different class with the i th sample, and we have $I = S + T + 1$. The supervised contrastive loss is calculated by the

following equation:

$$\begin{aligned} \gamma &= \sum_{t=1}^T \exp\left(\frac{\rho(\mathbf{z}_0^{(L)}(i), \mathbf{z}_0^{(L)}(t))}{\tau}\right) \\ &+ \sum_{s=1}^S \exp\left(\frac{\rho(\mathbf{z}_0^{(L)}(i), \mathbf{z}_0^{(L)}(s))}{\tau}\right) \\ \mathcal{L}_{\text{sup}}(\Theta) &= -\frac{1}{I} \sum_{i=1}^I \sum_{s=1}^S \log \frac{\exp(\rho(\mathbf{z}_0^{(L)}(i), \mathbf{z}_0^{(L)}(s))/\tau)}{\gamma} \end{aligned} \quad (7)$$

where $\rho(\mathbf{u}, \mathbf{v}) = (\mathbf{u} \cdot \mathbf{v}) / (\|\mathbf{u}\|_2 \|\mathbf{v}\|_2)$ measures the distance between two feature vectors. The supervised contrastive loss $\mathcal{L}_{\text{sup}}$ encourages the model to extract closely aligned features for all samples from the same class, thus improving the difference between the interclass features.

3) *Unsupervised Contrastive Loss*: The vectorized form collection of tokens $\mathbf{Z}^{(0)}$ consists of a class token $\mathbf{z}_0^{(0)}$ and data tokens $\{\mathbf{z}_n^{(0)}\}_{n=1}^N$. By fixing the class token and randomly shuffle the data tokens, we can obtain $\mathbf{Z}^{(0)} = [\mathbf{z}_0^{(0)}, \mathbf{z}_1^{(0)}, \dots, \mathbf{z}_N^{(0)}]$, where the values of $\{\mathbf{z}_n^{(0)}\}_{n=1}^N$ are shuffled data tokens. Then, we split the shuffled data tokens of $\mathbf{Z}^{(0)}$ into two groups, where the first group of data tokens with the class token are used to construct $\hat{\mathbf{Z}}^{(0)} = [\mathbf{z}_0^{(0)}, \mathbf{z}_1^{(0)}, \dots, \mathbf{z}_n^{(0)}]$ ($n \in [2, N-2]$), and the second group of data tokens with the class token are collected into $\check{\mathbf{Z}}^{(0)} = [\mathbf{z}_0^{(0)}, \mathbf{z}_{n+1}^{(0)}, \dots, \mathbf{z}_N^{(0)}]$. $\mathbf{Z}^{(0)}$, $\hat{\mathbf{Z}}^{(0)}$, and $\check{\mathbf{Z}}^{(0)}$ are input into transformer blocks to obtain the corresponding class tokens $\mathbf{z}_0^{(L)}$, $\hat{\mathbf{z}}_0^{(L)}$, and $\check{\mathbf{z}}_0^{(L)}$ in the L th block, respectively. The unsupervised contrastive loss is calculated as follows:

$$\mathcal{L}_{\text{uns}}(\Theta) = \left(\rho(\mathbf{z}_0^{(L)}, \hat{\mathbf{z}}_0^{(L)}) + \rho(\mathbf{z}_0^{(L)}, \check{\mathbf{z}}_0^{(L)}) - 2\right)^2. \quad (8)$$

The unsupervised loss term $\mathcal{L}_{\text{uns}}$ makes both $\hat{\mathbf{z}}_0^{(L)}$ and $\check{\mathbf{z}}_0^{(L)}$ similar to $\mathbf{C}$ by guiding the model to mine the relationship between tokens without relying on labels. In this way, the redundancy of tokens is reduced, and thus, the generalization of the model is improved.

4) *Regularization Term*: We introduce an ℓ_1 -regularizer on the MSA weights $\mathcal{A}$. The regularization term $\mathcal{L}_{\text{sp}}(\mathcal{A})$ can be calculated by the following equation:

$$\mathcal{L}_{\text{sp}}(\mathcal{A}) = \frac{1}{LM(N+1)^2} \|\mathcal{A}\|_1, \quad \|\mathcal{A}\|_1 = \sum_{\ell=1}^L \sum_{m=1}^M \sum_{a \in \mathbf{A}_m^{(\ell)}} |a| \quad (9)$$

where $\|\cdot\|_1$ represents the ℓ_1 -regularizer. The regularization term suppresses the fitting ability of the MSA, thus alleviating the over-fitting problem of the proposed CViT.

5) *Objective Function*: By combining the cross-entropy loss $\mathcal{L}_{\text{CE}}(\cdot)$ in (6), the supervised contrastive loss $\mathcal{L}_{\text{sup}}(\cdot)$ in (7), the unsupervised contrastive loss $\mathcal{L}_{\text{uns}}(\cdot)$ in (8), and the regularization term $\mathcal{L}_{\text{sp}}(\cdot)$ in (9), we consider the following optimization problem:

$$\min_{\Theta, \mathcal{A}} \mathcal{L}_{\text{CE}}(\Theta) + \lambda_1 \mathcal{L}_{\text{sup}}(\Theta) + \lambda_2 \mathcal{L}_{\text{uns}}(\Theta) + \lambda_3 \mathcal{L}_{\text{sp}}(\mathcal{A}) \quad (10)$$

where λ_1 – λ_3 are utilized to balance the different losses.

TABLE II
DETAILS OF THE SA, PU, AND HC HSI DATASETS

	Resolution	Classes	Labeled samples
SA	$208 \times 512 \times 217$	16	54129
PU	$103 \times 610 \times 340$	9	42776
HC	$274 \times 1217 \times 303$	16	257530

III. EXPERIMENTS

A. Experimental Configuration and Datasets

A series of experiments are conducted on three widely used HSI datasets: Salinas (SA) [19], Pavia University (PU) [1], and WHU-Hi-HanChuan (HC) [20]. Table II lists their detailed information, including the resolution, number of classes, and number of labeled samples. Four related deep learning methods are adopted for comparison: ConvNeXt [21], Swin transformer (Swin) [14], ConvViT [10], and ViT [9]. Swin, ConvViT, and ViT are transformer-based methods, and ConvNeXt has the same macro architecture as the proposed CViT. All the methods are optimized with the AdamW optimizer on a PC platform with RTX 3080 and trained for 200 epochs. The learning rate is scaled by a cosine decay learning rate scheduler. The grid search method is applied to obtain the hyperparameters of the CViT. The learning rate is 0.001, the batch size $I = 64$, the patch size $P = 3$, the dimension of tokens $D = 32$, the number of multiheads $M = 12$, the number of transformer blocks $L = 2$, the temperature parameter $\tau = (D)^{1/2}$, and the balance parameters $\lambda_1 = \lambda_2 = \lambda_3 = 1$.

B. Classification Results With 1.5% Training Samples

In this section, we randomly select 1.5% labeled samples per class for training and the rest labeled samples for testing. The overall accuracy (OA), average accuracy (AA), kappa coefficient ($\mathcal{K}$), training time for 200 epochs (training), testing time for all labeled samples (testing), and the number of parameters (Params.) are employed to analyze the performance of different methods quantitatively.

Table III reports the comparison of the classification performance of our CViT with other methods. We can observe that although CViT takes a long time for training, it has the same testing time as ViT and higher accuracy. This phenomenon is due to the use of contrastive learning when training CViT, but no longer required for testing.

Fig. 2 visualizes the classification maps of all considered methods for the SA dataset. CViT is more discriminative for confusion regions, which illustrates the effectiveness of the proposed modules. Swin mitigates the over-smoothing issue of global MSA by local shift windows, while CViT alleviates the over-smoothing issue by introducing contrastive learning and ℓ_1 -regularizer.

Fig. 3 presents the training and testing OA curves of different methods. The training OA of all competitors is higher than the testing OA, and the training OA of CViT is more similar to the testing OA. This suggests that CViT is not prone to over-fitting but may require more epochs to train the model.

TABLE III

CLASSIFICATION RESULTS FOR SA [19], PU [1], AND HC [20] HSI DATASETS USING 1.5% OF THE LABELED SAMPLES FOR TRAINING AND 15×15 INPUT SIZE (MEAN, AVERAGED OVER FIVE RUNS)

		ConvNeXt [21]	Swin [14]	ConvViT [10]	ViT [9]	CViT
SA	OA (%)	97.53	97.68	97.04	96.00	97.88
	AA (%)	98.89	98.97	98.34	97.79	98.77
	$\mathcal{K} \times 100$	97.26	97.42	96.71	95.56	97.64
	Training (s)	828.50	709.95	585.20	712.12	863.98
	Testing (s)	28.06	25.03	25.12	43.36	44.73
	Params. (K)	329.49	321.55	180.02	261.55	261.55
PU	OA (%)	96.69	96.87	97.08	95.75	98.23
	AA (%)	92.86	93.08	93.98	91.66	96.58
	$\mathcal{K} \times 100$	95.61	95.85	96.12	94.35	97.65
	Training (s)	448.24	341.39	331.69	403.38	487.22
	Testing (s)	38.50	24.78	27.63	64.35	64.47
	Params. (K)	268.11	260.17	149.55	231.08	231.08
HC	OA (%)	96.51	97.79	98.22	97.53	98.77
	AA (%)	93.04	95.94	97.04	95.11	97.81
	$\mathcal{K} \times 100$	95.91	97.42	97.92	97.11	98.57
	Training (s)	4020.93	4022.54	3587.55	4082.01	4899.93
	Testing (s)	106.61	101.89	98.40	166.90	169.82
	Params. (K)	367.50	359.57	199.02	280.56	280.56

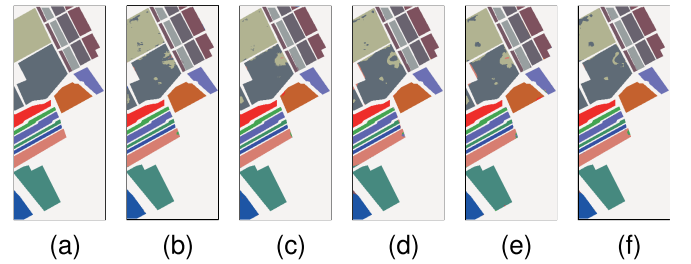


Fig. 2. Classification maps with 1.5% training samples and 15×15 input size for the SA dataset by different methods. (a) Ground truth. (b) ConvNeXt (97.64%). (c) Swin (97.92%). (d) ConvViT (97.21%). (e) ViT (96.38%). (f) CViT (98.07%).

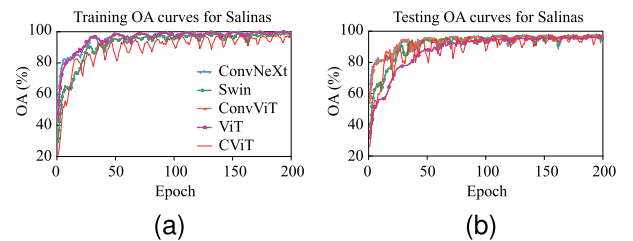


Fig. 3. Training and testing OA curves of different methods using 1.5% of the labeled samples for training and 15×15 input size. (a) SA. (b) PU.

C. Compared With Different Training Samples

In this section, we randomly select 1.0%, 1.5%, 2.0%, 2.5%, and 3.0% labeled samples of SA and PU datasets per class for training, as shown in Fig. 4. CViT has more advantages when using 1% of the training samples, while the accuracy of the different methods is more similar when using 3% of the training samples. This indicates the strong performance of CViT with limited training samples.

D. Ablation Study

In order to verify the effect of contrastive losses and regularization terms, Fig. 5 shows the comparison of the

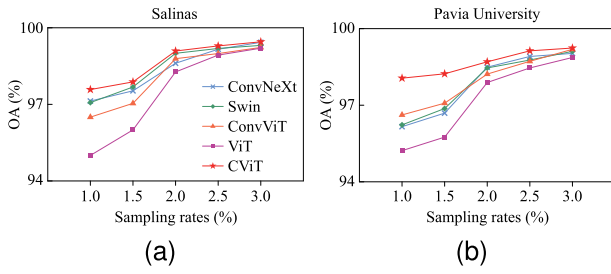


Fig. 4. OA of different methods using different training samples and 15×15 input size. (a) SA. (b) PU.

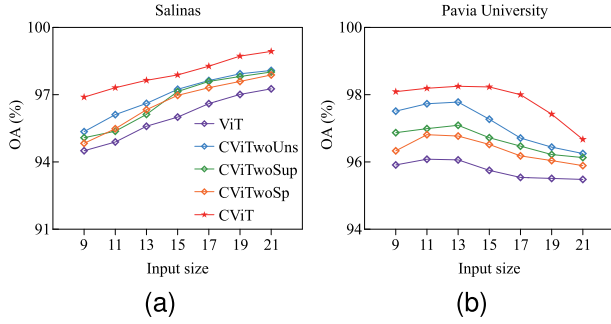


Fig. 5. OA of different methods using 1.5% of the labeled samples for training and different input sizes. (a) SA. (b) PU.

OA with different input sizes of $H \times W$ ($H = W$, $H \in \{9, 11, 13, 15, 17, 19, 21\}$) and different methods. The methods used for comparison include CViT, CViT without $\mathcal{L}_{sp}$ (CViT-woSp), CViT without $\mathcal{L}_{sup}$ (CViTTwoSup), CViT without $\mathcal{L}_{uns}$ (CViTTwoUns), and original ViT.

Fig. 5 presents that the CViT outperforms all the competitors, exhibiting the effectiveness of loss and regularization terms. Moreover, larger input sizes can provide more long-range dependencies and may exacerbate the over-smoothing issue caused by the MSA. The results of the ablation study show that the contrastive losses and regularization terms can effectively improve the classification accuracy and generalizability of the method.

IV. CONCLUSION AND DISCUSSION

In this letter, we propose CViT to provide a practical way to train a model with high accuracy and generalization for HSI classification. We alleviate the over-smoothing issue caused by the MSA by linear weights and ℓ_1 -regularizer. Moreover, we introduce additional training targets where the contrastive losses guide the model to mine more information from the data and alleviate the over-fitting issue. Contrastive learning significantly improves the classification accuracy and generalization of the model. Experimental results on three HSI datasets show that CViT outperforms all competitors with limited training samples. In future work, the proposed method can be extended to perform for weakly supervised and unsupervised HSI classification.

REFERENCES

- [1] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral-spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2017.
- [2] K. Jiang, Z. Wang, P. Yi, G. Wang, T. Lu, and J. Jiang, "Edge-enhanced GAN for remote sensing image superresolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5799–5812, Jun. 2019.
- [3] Y. Xiao, X. Su, Q. Yuan, D. Liu, H. Shen, and L. Zhang, "Satellite video super-resolution via multiscale deformable convolution alignment and temporal grouping projection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5610819.
- [4] Y. Xiao et al., "Space-time super-resolution for satellite video: A joint framework based on multi-scale spatial-temporal transformer," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 108, Apr. 2022, Art. no. 102731.
- [5] H. Zhou, C. Zhang, X. Zhang, and Q. Ma, "Image classification based on quaternion-valued capsule network," *Appl. Intell.*, vol. 53, pp. 5587–5606, Mar. 2023.
- [6] S. Jia, S. Jiang, Z. Lin, N. Li, M. Xu, and S. Yu, "A survey: Deep learning for hyperspectral image classification with few labeled samples," *Neurocomputing*, vol. 448, pp. 179–204, Aug. 2021.
- [7] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5518615.
- [8] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, Dec. 2017, pp. 5998–6008.
- [9] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. 9th Int. Conf. Learn. Represent.*, May 2021, pp. 1–21.
- [10] Z. Zhao, D. Hu, H. Wang, and X. Yu, "Convolutional transformer network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [11] X. Huang, M. Dong, J. Li, and X. Guo, "A 3-D-swin transformer-based hierarchical contrastive learning method for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5411415.
- [12] Y. Dong, J. Cordonnier, and A. Loukas, "Attention is not all you need: Pure attention loses rank doubly exponentially with depth," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, Jul. 2021, pp. 2793–2803.
- [13] Z. Chen, L. Xie, J. Niu, X. Liu, L. Wei, and Q. Tian, "Visformer: The vision-friendly transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 569–578.
- [14] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vision. (ICCV)*, Oct. 2021, pp. 9992–10002.
- [15] L. Zhao, W. Luo, Q. Liao, S. Chen, and J. Wu, "Hyperspectral image classification with contrastive self-supervised learning under limited labeled samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [16] J. Li, X. Li, Z. Cao, and L. Zhao, "ROBYOL: Random-occlusion-based BYOL for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [17] P. Khosla et al., "Supervised contrastive learning," in *Proc. Adv. Neural Inf. Process. Syst.*, Dec. 2020, pp. 1–13.
- [18] K. He, H. Fan, Y. Wu, S. Xie, and R. B. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735.
- [19] G. Vane, R. O. Green, T. G. Chrien, H. T. Enmark, E. G. Hansen, and W. M. Porter, "The airborne visible/infrared imaging spectrometer (AVIRIS)," *Remote Sens. Environ.*, vol. 44, nos. 2–3, pp. 127–143, 1993.
- [20] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-Hi: UAV-borne hyperspectral with high spatial resolution (H^2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.
- [21] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11966–11976.